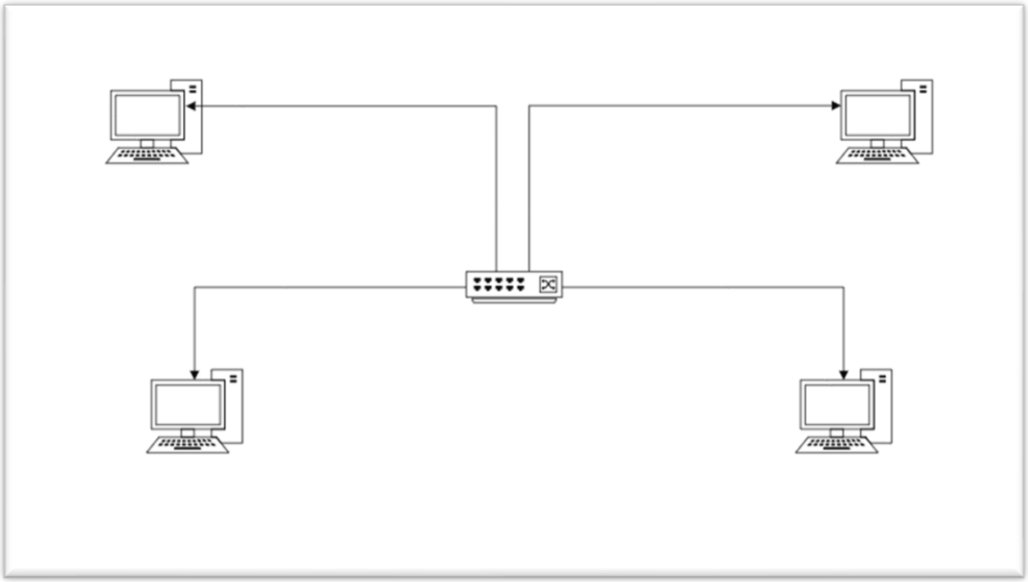
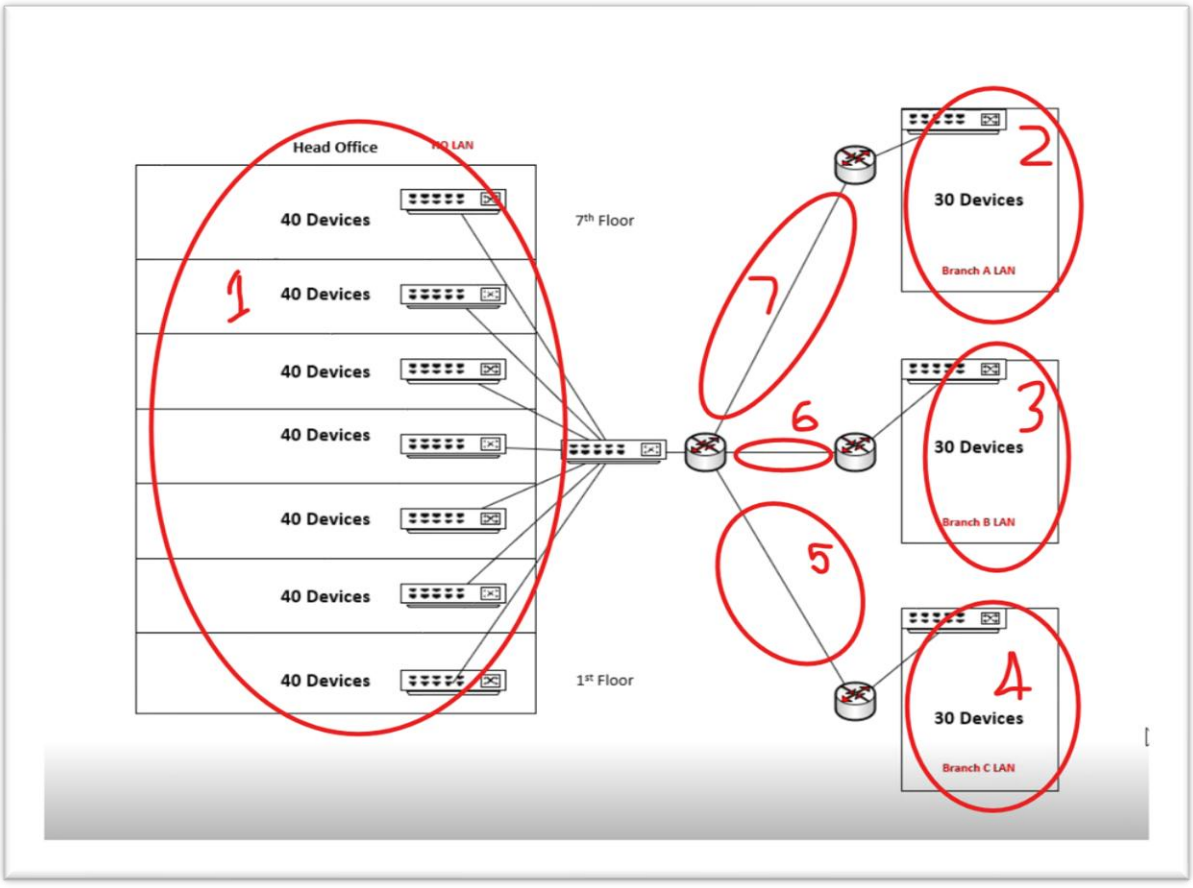


WAN



Building a Company Network (Head Office and Branches)

Suppose we want to build a network for a company. The company has a head office and other branches, and we want to connect them all together.

Step 1: Connect computers using switches

The head office has several floors. On each floor, we connect the computers to each other using a switch. After that, all the floor switches are connected to one main switch.

Step 2: Connect the head office with the branches

Now we need to connect the head office and the other branches together. To do this, we use a router, because a router is used to connect different networks.

For example, let's say the company has 7 networks in total. If network 2 wants to connect with network 7, we need a router. If network 7 wants to connect with network 1, we need another router. In this way, we may need up to 4 routers to connect all the branches with the head office.

However, if the head office and the branches are all located in the same building, we can use VLAN technology instead. In that case, we only need one router.

Step 3: How to connect the routers

There are a few simple ways to connect routers:

1. Using cables through an ISP (Internet Service Provider), such as SLT or Dialog.
    - How it works: One router connects to the ISP's circuit board. This circuit board connects to towers, and the towers communicate with other towers. This is how the routers get connected to each other.
2. Using a wireless connection between routers.
3. Using a VPN tunnel to connect the routers securely.

How to Connect Routers (Simple Explanation)

There are a few simple ways to connect routers:

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Hyper-V

What is Hyper-V?
Hyper-V is a tool used to create virtual machines.

Difference Between VMware and Hyper-V

Both VMware and Hyper-V are used to create virtual machines, but they work in a different way.

Hyper-V connects directly to the hardware of the host machine. VMware connects to the Windows OS of the host machine first, and then the Windows OS connects to the hardware.

Why does this matter?

Hyper-V does not depend on the host OS to run. This means if we want to update Windows while Hyper-V is running, it is not a problem. But if we are using VMware, we usually need to turn off the virtual machines first.

Since Hyper-V connects directly to the hardware, it gives better performance.

Also, if we use a Linux server on Hyper-V, we get good performance. But if we use a Linux GUI (desktop version), the performance is not as good.

Types of Virtual Switches in Hyper-V

1. External Virtual Switch

- Purpose: Connects a virtual machine to the physical network and the internet.
- How it works: It links to a physical network adapter on the computer.
- Best for: Virtual machines that need internet access, need to be reached by other physical computers in the office, or need an IP address from the physical router.

2. Internal Virtual Switch

- Purpose: Allows communication between the host computer and the virtual machines.
- How it works: It does not link to a physical network card. It creates a private network that only the host and the VMs on that switch can use.
- Best for: Building a lab where you want to manage your VMs from your host computer, but keep them separate from the outside internet.

(Think of it like the lobby of a building. The security guard, which is the host OS, can talk to the office workers, which are the VMs. But there is no door leading to the street.)

3. Private Virtual Switch

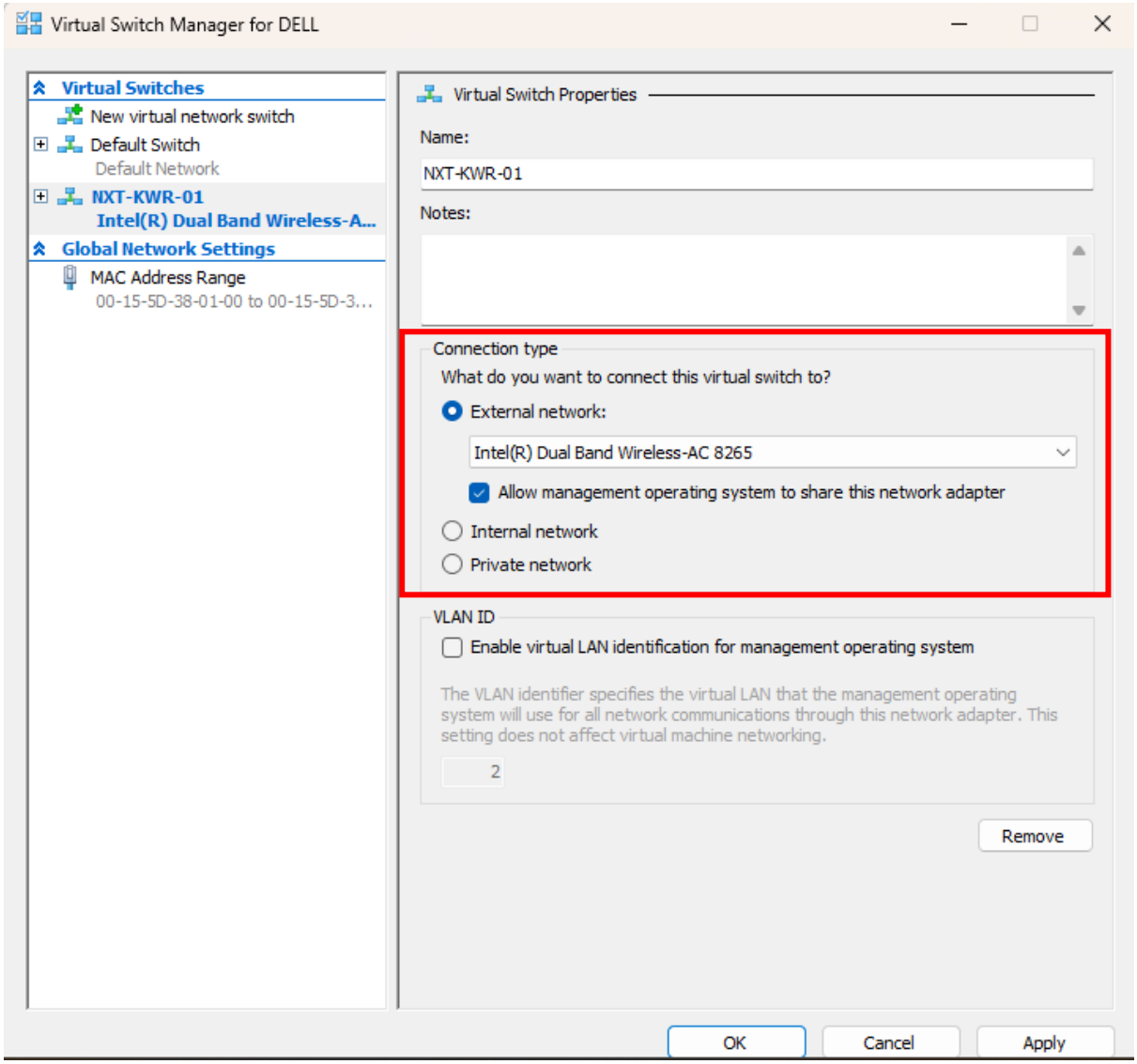
Purpose: Allows communication only between virtual machines.

How it works: The host OS cannot see this network, and it has no connection to the physical network at all.

Best for: Highly secure or isolated testing, where VMs only need to talk to each other.

It is 100% isolated, making it the safest place to test something dangerous, like a virus, because it cannot spread to the host or the outside network. (For example: testing malware or a private database cluster.)

When we create virtual switch



When we select the External Switch, we need to choose either **Wi-Fi or Ethernet.**

- If the host computer uses Wi-Fi to connect to the internet, we should select Wi-Fi.
- If the host computer uses Ethernet to connect to the internet, we should select Ethernet.

There is also a tickbox option:

- If we untick the box, **only the VMs get internet access**, not the host.
- If we tick the box, **both the host and the VMs get internet access.**

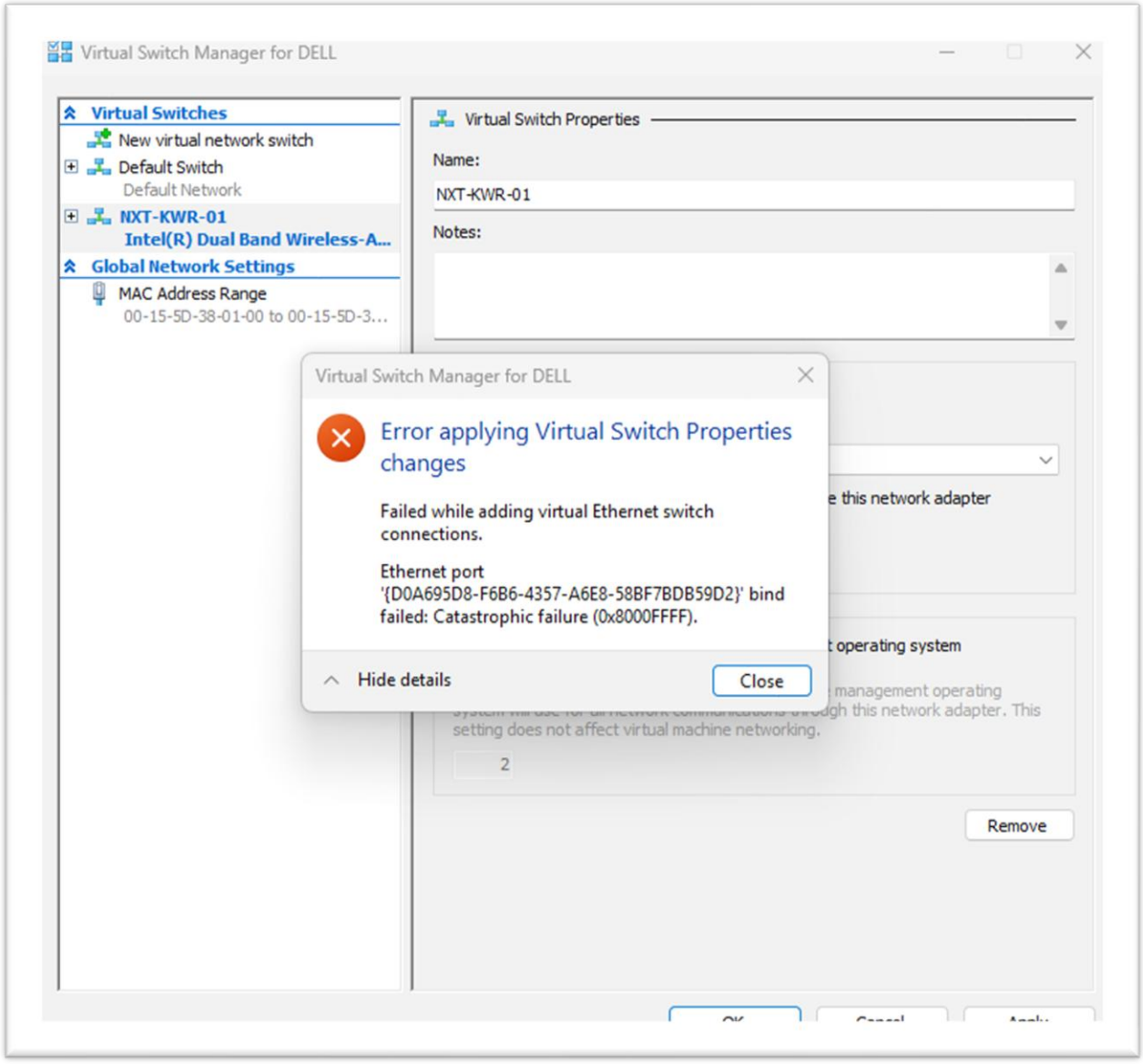
VLAN ID

Imagine a Hyper-V switch has several VLANs, for example VLAN 10 and VLAN 20. If we want to access VLAN 10, how do we do it?

To access a specific VLAN, we need to know its VLAN ID. We tick the VLAN box and type in the VLAN number. This lets us access that VLAN.

When we are accessing a VLAN, the host's Wi-Fi network connection can be turned off. This way, we can send data through the Ethernet cable, while still using Wi-Fi as the host computer's own network connection.

Troubleshoot



Error: Catastrophic Failure (0x8000FFFF) When Creating External Virtual Switch

Sometimes, when we try to create an External Virtual Switch in Hyper-V using a Wi-Fi adapter, we get an error message that says "Catastrophic failure (0x8000FFFF)". This error means the system failed to bind the virtual switch to the physical network adapter.

Common causes and fixes:

1. Wireless adapter driver issue - This is the most common cause. Updating the Wi-Fi adapter driver through Device Manager often solves the problem.

2. Adapter binding stuck - Disabling and then re-enabling the Wi-Fi adapter can reset its connection state and clear the error.
3. Hyper-V service issue - Restarting the "Hyper-V Virtual Machine Management" service (through the Services window) can fix a stuck virtual switch creation process.
4. Antivirus or firewall interference - Some antivirus programs block changes to network adapter bindings. Turning off third-party antivirus temporarily can help identify this.
5. Using PowerShell instead of GUI - Sometimes creating the switch through PowerShell command works even when the graphical interface fails.
6. System restart - If nothing else works, restarting the computer often clears low-level network errors like this one.

Step by step

Disable then re-enable the wireless adapter

Go to Control Panel > Network Connections. Right-click the Wi-Fi adapter (Intel Dual Band Wireless), select Disable, wait a few seconds, then Enable it again. This resets the adapter's binding state, which often clears the catastrophic failure error.

Update the wireless adapter driver

Go to Device Manager > Network Adapters > right-click the Intel Dual Band Wireless adapter > Update Driver. Old or buggy Wi-Fi drivers are the most common cause of this exact error when creating an External Virtual Switch on a wireless card.

Restart the Hyper-V Virtual Machine Management service

Open Services (services.msc), find 'Hyper-V Virtual Machine Management', right-click and select Restart. This service controls virtual switch creation, and restarting it can clear a stuck binding.

Check if antivirus or firewall is blocking the bind

Some antivirus software blocks changes to network adapter bindings. Temporarily disable third-party antivirus/firewall software and try creating the External Switch again.

Try creating the switch via PowerShell instead

Open PowerShell as Administrator and run: New-VMSwitch -Name "NXT-KWR-01" -NetAdapterName "Wi-Fi" -AllowManagementOS $true. PowerShell sometimes succeeds where the GUI fails, and it gives more detailed error output if it fails again.

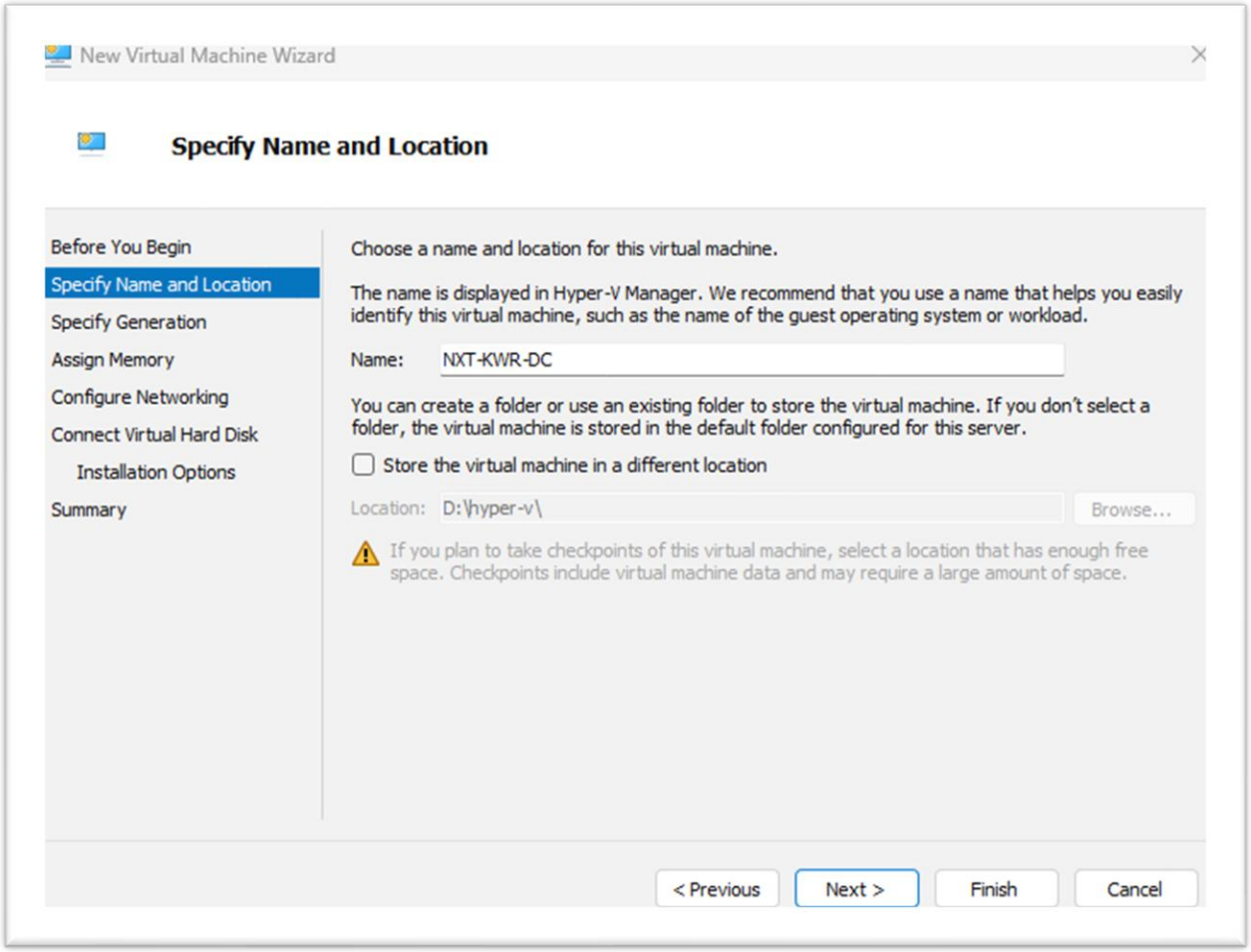
Restart the computer

If none of the above works, a full restart clears many low-level network binding issues. After restart, try creating the External Virtual Switch again before doing anything else

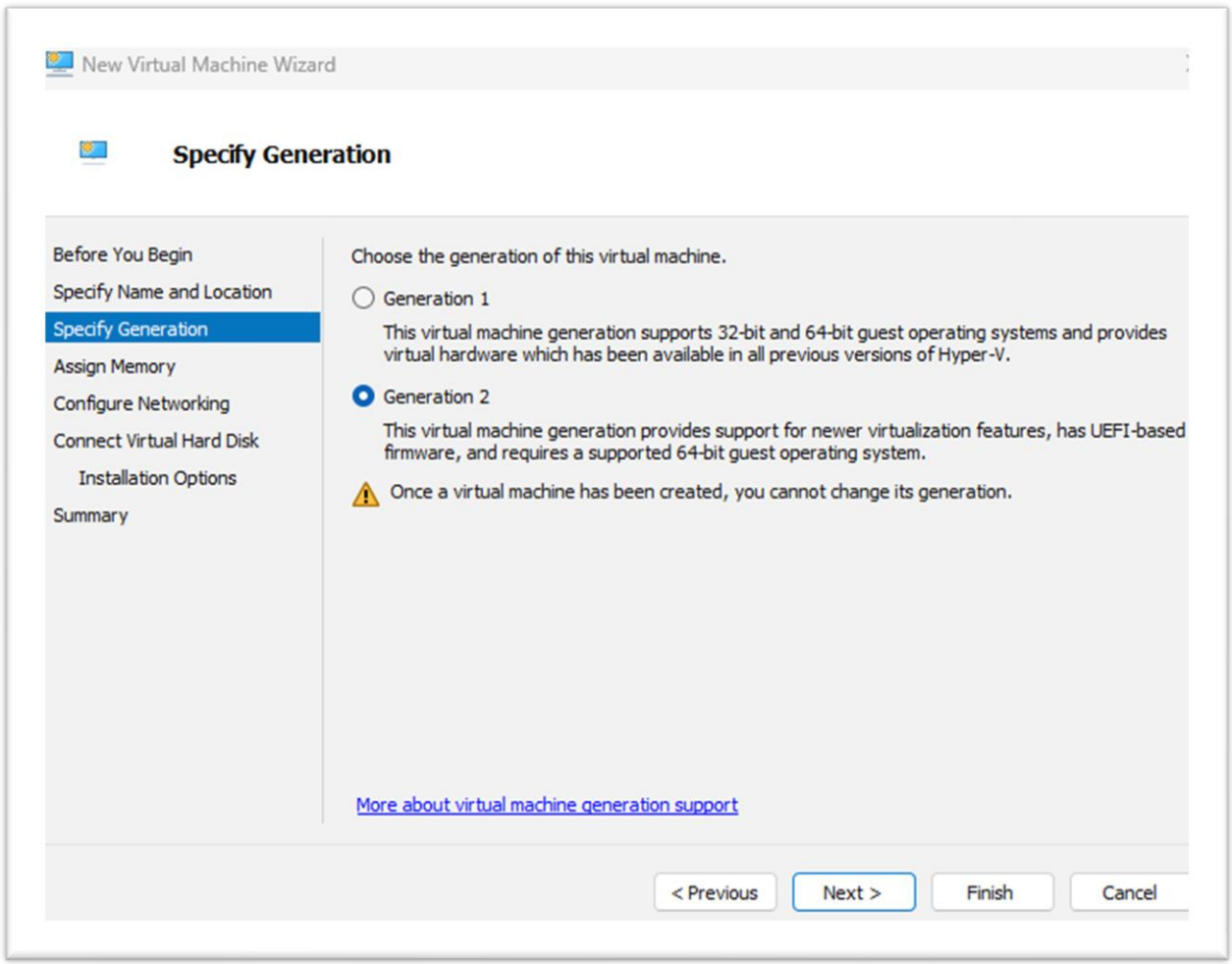
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Create Virtual Machine on Hyper V

Step 1: Give a name
Give any name to the virtual machine.



Step 2: Choose the generation



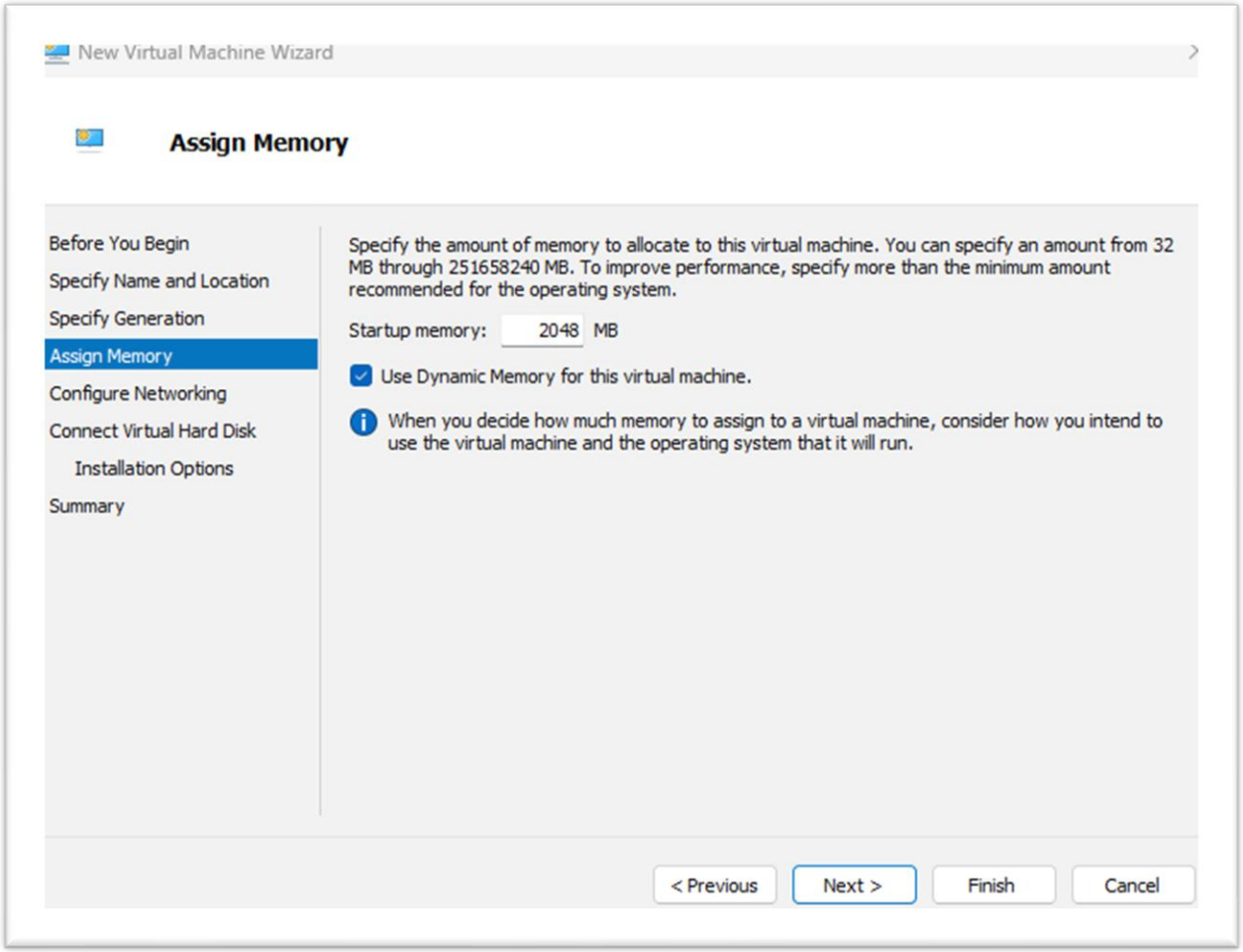
Generation 1

- Supports older operating systems.
- Uses IDE controllers for hard drives, which is slower and older technology.
- Supports both 32-bit and 64-bit operating systems.
- Use this only for very old operating systems, like Windows XP, Windows 7 (32-bit), or old Linux versions.

Generation 2

- Designed for modern, 64-bit operating systems.
- Supports Secure Boot and Virtual TPM. This is required if you want to run Windows 11 as a VM.
- Only supports 64-bit operating systems.
- Always use this for Windows 10, Windows 11, Windows Server 2025, and modern 64-bit Linux systems.

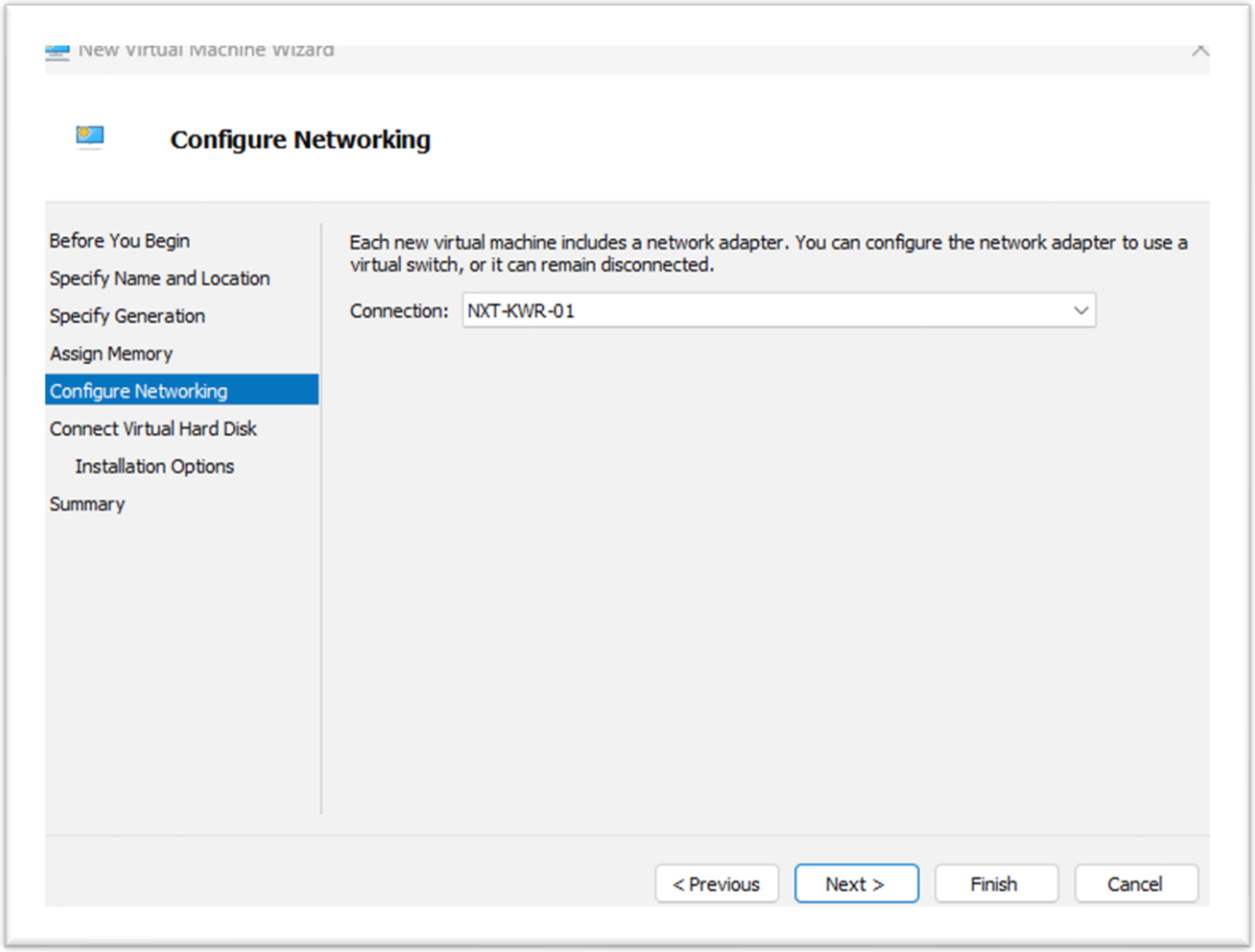
Step 3: Assign memory



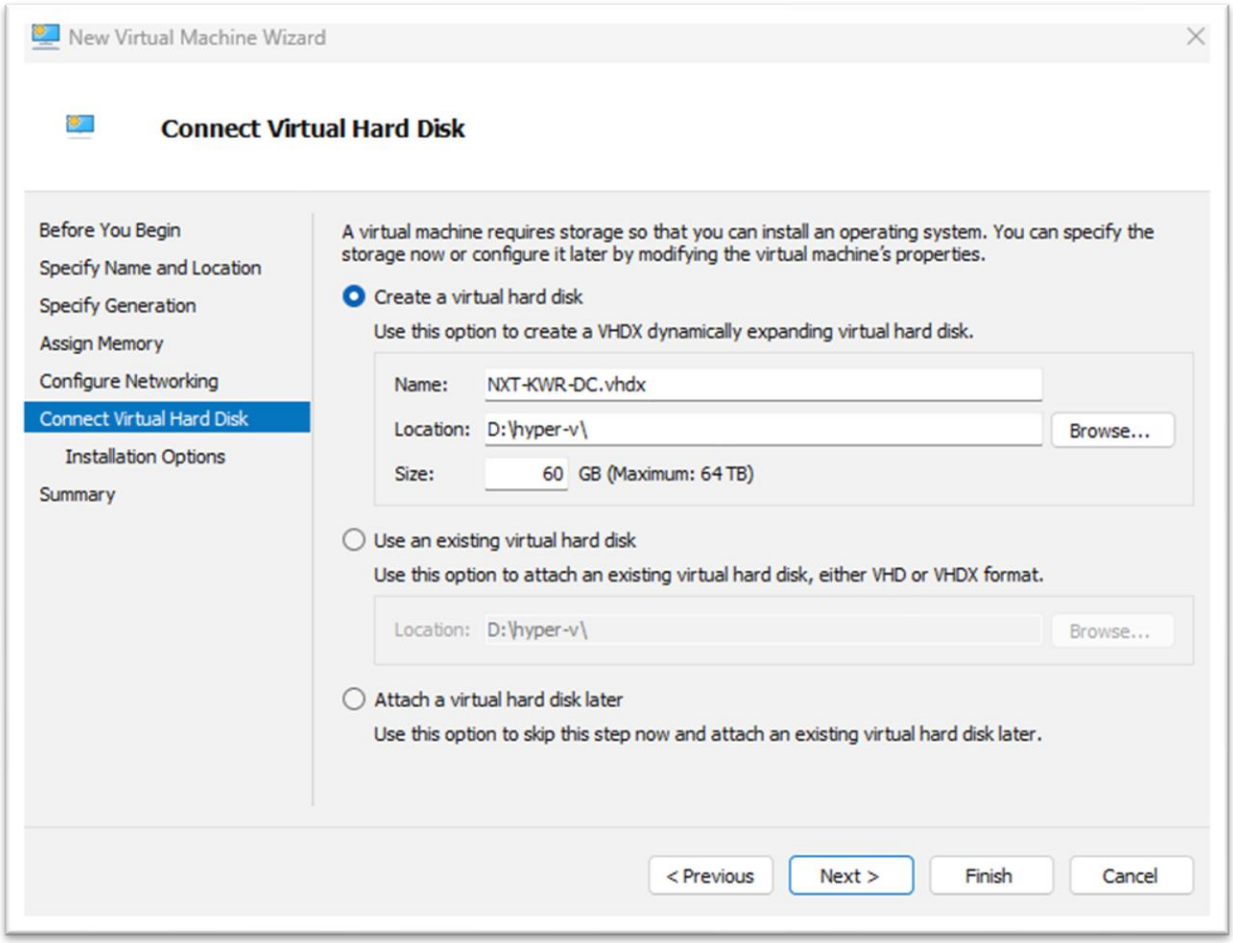
- **Startup Memory:** The amount of RAM given to the VM when it starts.
- **Dynamic Memory:** If we turn this on, Hyper-V automatically adjusts the RAM based on what the VM actually needs. This saves physical RAM for other tasks, and it is helpful when running multiple VMs. If we turn this off, the VM takes a fixed amount of RAM (for example, 2GB) and keeps it until it is shut down.

Step 4: Configure networking

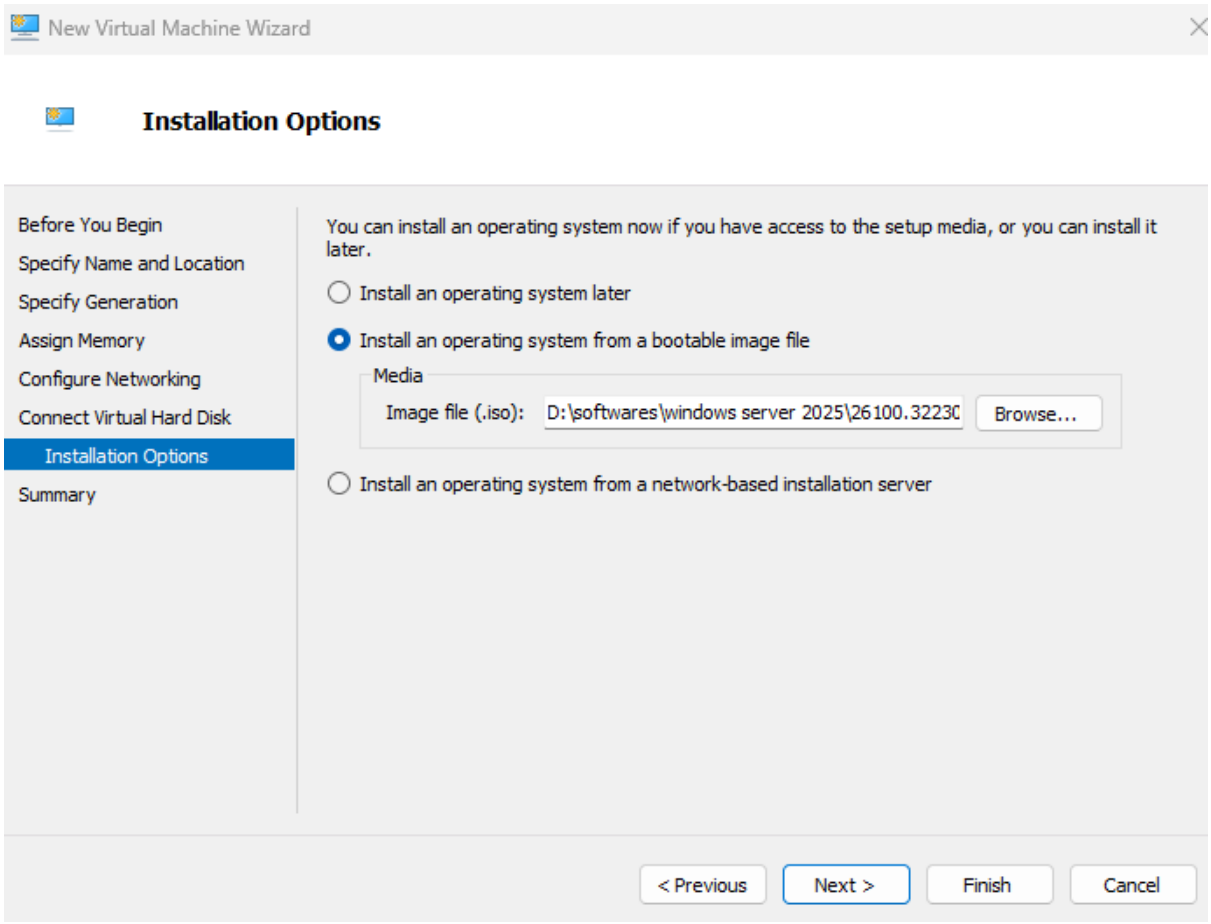
Choose the network adapter type: External, Internal, or Private.



Step 4: Connect virtual hard disk



Step 5: Installation Options

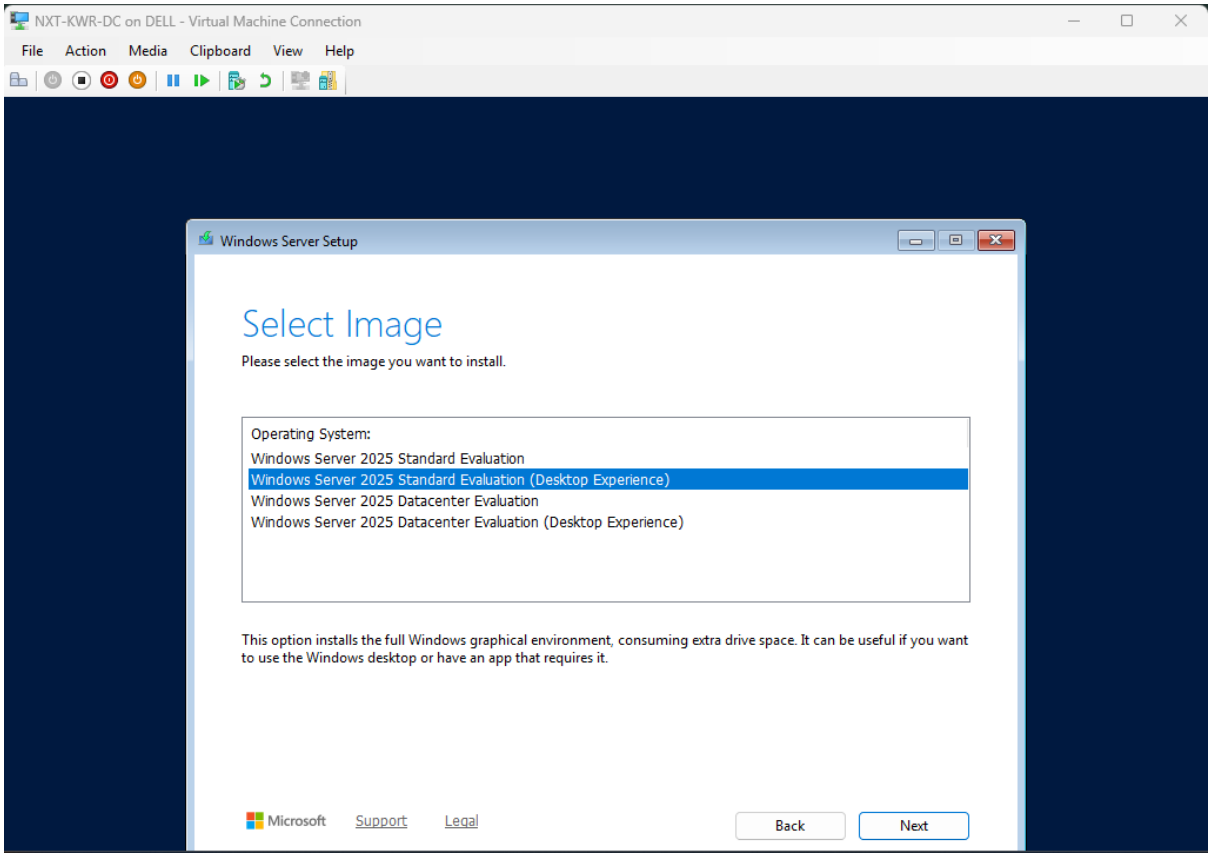


Server Operating System

If we install a server operating system on any device, that device becomes a server, no matter what hardware it has. For example, if we install Windows Server on a laptop or PC, it becomes a server.

A normal Windows operating system is used for gaming, office work, and everyday tasks. A server operating system is used to create servers, such as Active Directory or a file server.

When Installing Windows Server 2025 on a VM



What is the difference between Standard Evaluation and Datacenter Evaluation?

- **Standard Evaluation:** With one license, we can only run 1 or 2 VMs (servers).
- **Datacenter Evaluation:** With one license, we can run unlimited VMs.

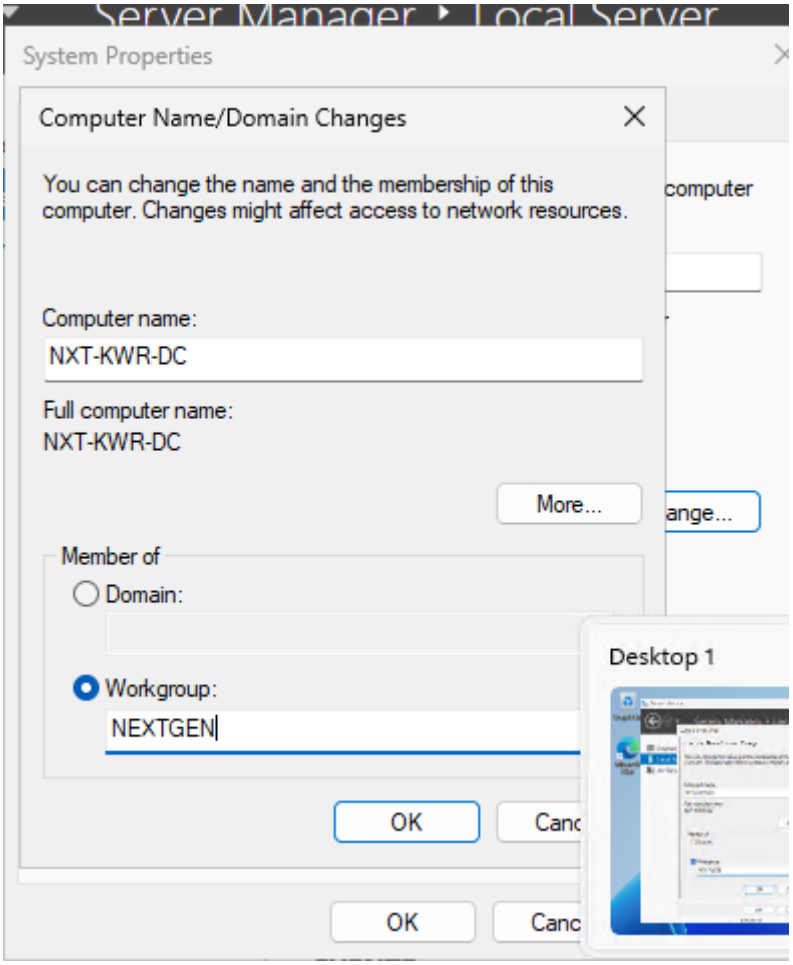
Desktop Experience option:

- **With Desktop Experience:** The OS has a GUI (graphical interface).
- **Without Desktop Experience:** The OS has no GUI, and only works using command lines.

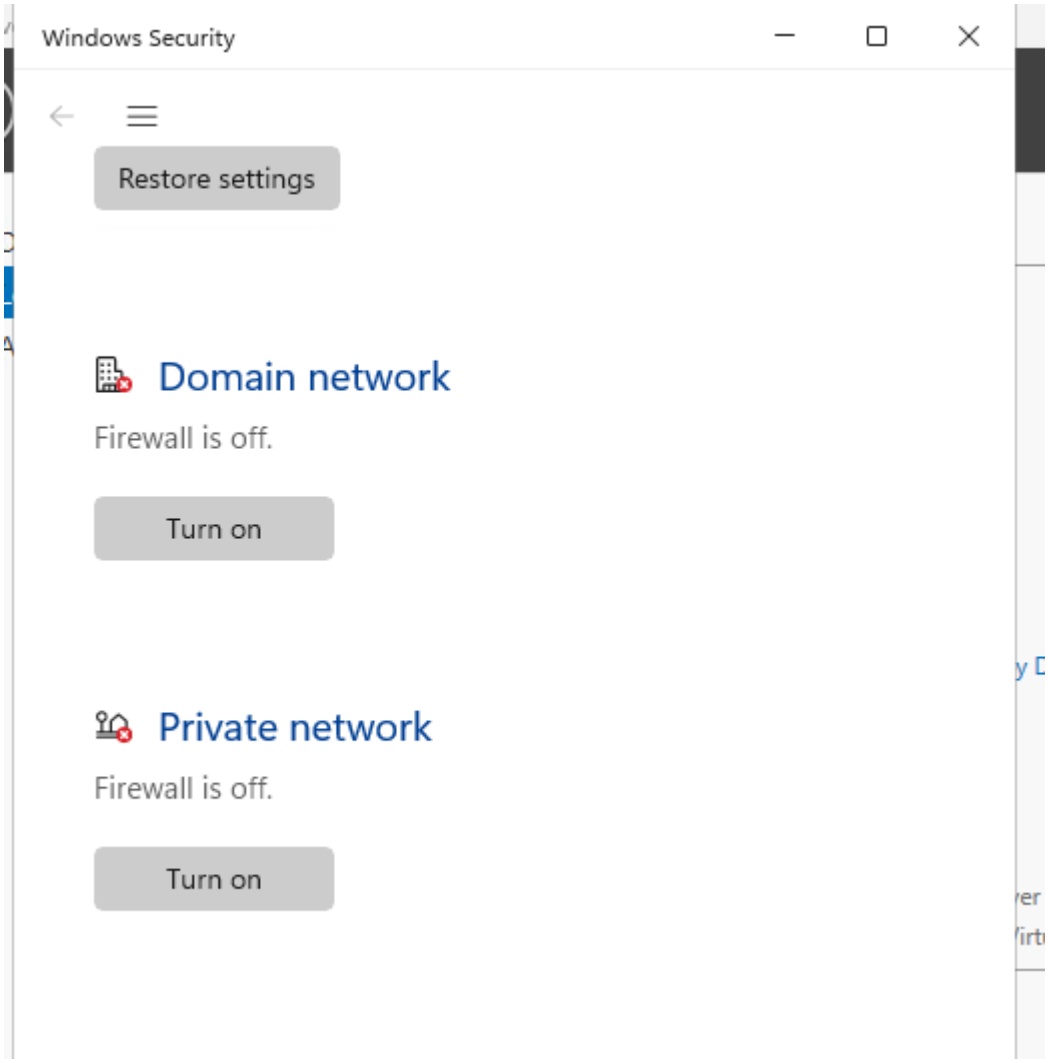
Very Important: Best Practices Before Setting Up the Server

When we install a server (Windows Server 2025) on our VM, there are some important first steps we should follow:

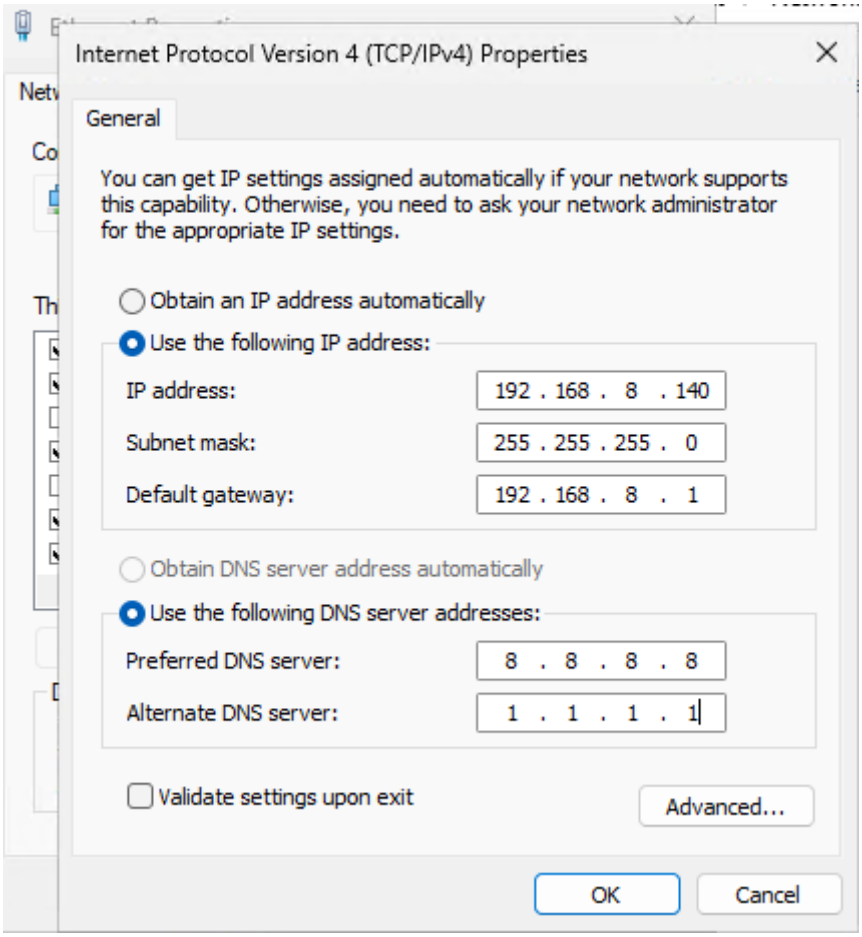
1. **Change the server name and workgroup.**



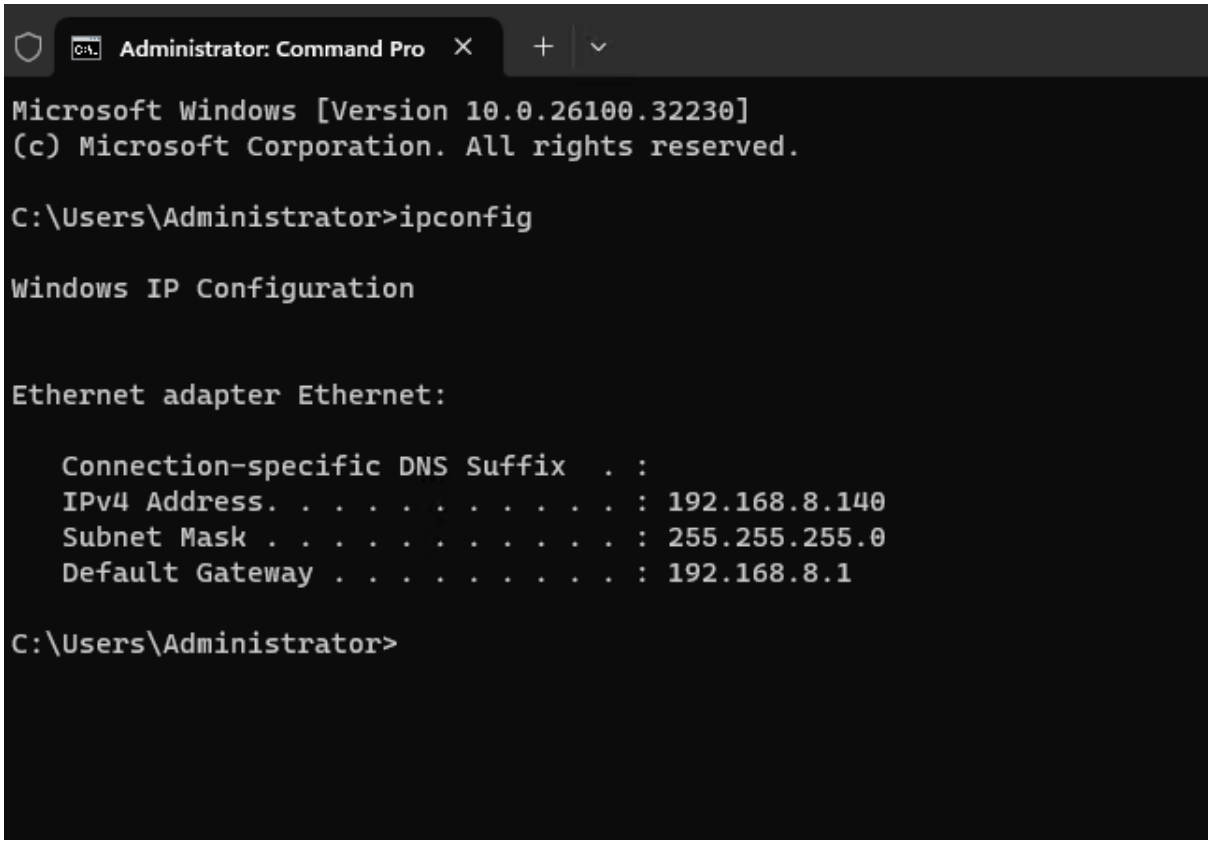
2. **Turn off the firewall.**



3. **Enable Remote Management.**
4. **Enable Remote Desktop.**
5. **Configure the Ethernet settings:**
    - First, turn off IPv6. As beginners, this helps us avoid unnecessary problems.
    - For IPv4, assign a static IP address (instead of an automatic one).



When we give IP address we want to check our host IP address



What Happens When You Create an External Switch in Hyper-V

When we create an External Switch in Hyper-V, Windows transfers control of the physical Wi-Fi card to Hyper-V. This means the Wi-Fi card is no longer directly connected to Windows.

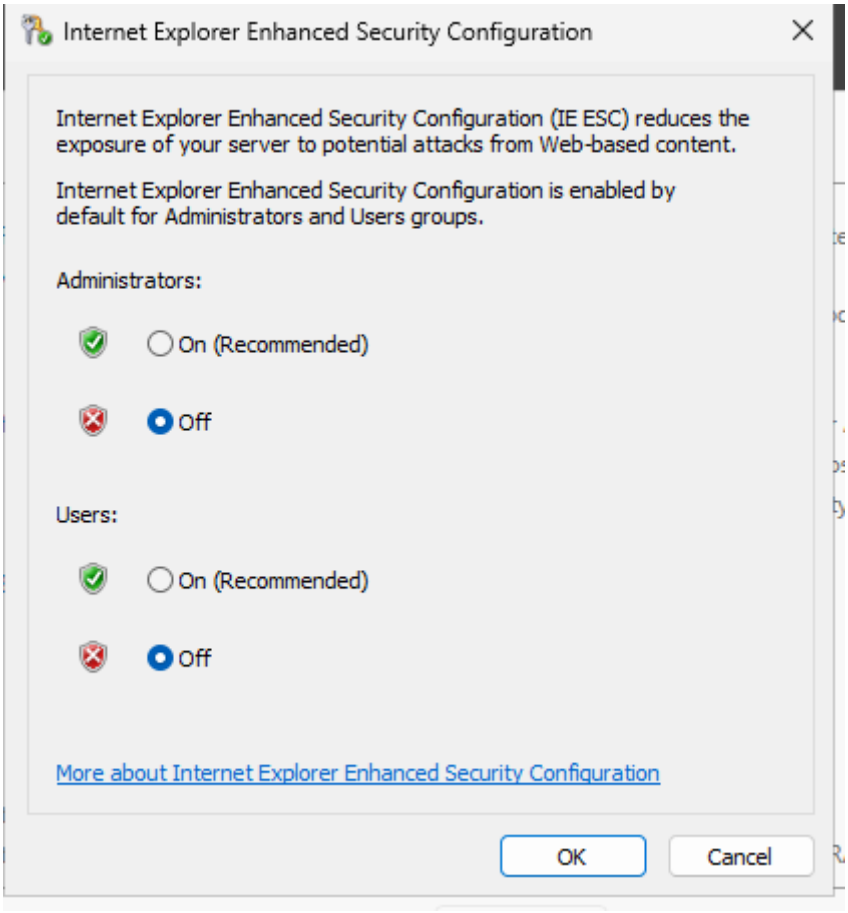
Instead, a Virtual Switch (for example, NXT-ESW-01) is created using the Wi-Fi card. Now, the laptop (the host computer) accesses the internet through this Virtual Switch. Because of this, we will see "vEthernet" instead of "Wi-Fi" in the network settings.

Example IP details:

- Host computer IP: 192.168.8.140 (this IP is given to the host computer by the router)
- Default gateway: 192.168.8.1

Other Important Settings

1. **Remote SSH Access:** When we use a Linux server, we turn on this feature so we can access it remotely, similar to Remote Desktop.
2. **Product ID:** At the enterprise level, this is used to activate the operating system.
3. **Time Zone:** All devices in the network should be set to the same time zone. This is especially important for digital forensics and other tasks, where accurate timestamps matter.
4. **IE Enhanced Security Configuration:** (as explained before, usually turned off in lab/practice environments for easier browsing)



What Happens If IE Enhanced Security Configuration Is Enabled?

If this setting is enabled, every time we visit a website from the server, a message will pop up asking "Is this safe?" It will also block scripts and images on many websites. This is done to protect the server from viruses or malware.

What Happens If It Is Disabled?

If we disable it, we can visit any website without problems, just like on a normal laptop.

Note: In an enterprise network (real production environment), this setting should be enabled for better security.

For the other settings, we can keep them as default.

Final Step: Update the OS

At the end, we must update our Windows Server OS to get the latest updates and security patches.

After updating, **restart the server.**

What is a Checkpoint?

A checkpoint is like an "Undo" button or a "Save Point" for a virtual machine.

Purpose:
It saves the exact state of the VM (its files, settings, and applications) at a specific point in time.

Why use it?
If we make a mistake, or something breaks the server, we can "apply" the checkpoint to go back to the point when the server was working correctly.

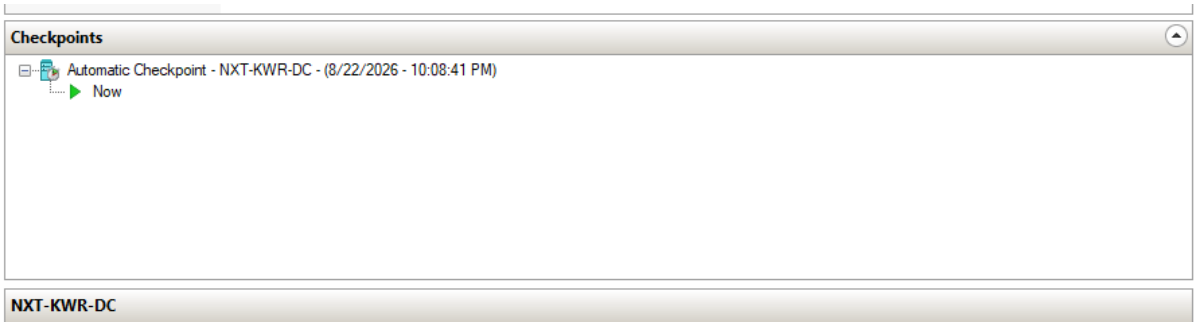
Benefit for Lab Practice:
As a student, it is a good habit to take a checkpoint before installing any new role (like DNS or Active Directory). If something goes wrong, we don't need to delete the whole VM and start over. We can just go back to the checkpoint.

For example: Before doing anything risky, like installing Active Directory or changing the IP address, we should take a checkpoint. If the server has a problem afterward, we can restore it back to its earlier working state with just one click.

What Gets Saved in a Checkpoint?

- The data on the server's hard drive at that point in time.
- The settings and hardware configuration of the server at that point in time.
- Anything currently running in the RAM (memory) of the server.

Important Advantage:
One of the biggest advantages of Hyper-V is that we can take a checkpoint while the virtual machine is still running (live), without any interruption to its work.



OSI layer / IP addressing / subnetting / NAT

Types of Networks

Type	Description	Example
LAN	Local Area Network	Office, school lab
MAN	Metropolitan Area Network	City-wide campus
WAN	Wide Area Network	Internet
PAN	Personal Area Network	Bluetooth between phone and headset

LAN (Local Area Network): A LAN connects computers and devices that are close to each other, usually within one building. For example, all the computers in an office or a school lab are connected using a LAN.